

Title: Machine Learning Approaches for E-Commerce Sales Forecasting

Domain

E-Commerce | Machine Learning | Predictive Analytics

Dataset

I recommend using three benchmark datasets together:

1. Rossmann Store Sales Dataset (Kaggle) (Daily retail sales, promotions, holidays, and customer information)

<https://www.kaggle.com/competitions/rossmann-store-sales>

2. Online Retail II Dataset (UCI Machine Learning Repository) (Transaction-level online retail data including products, quantities, prices, and customer information)

<https://archive.ics.uci.edu/dataset/502/online+retail+ii>

3. Brazilian E-Commerce Public Dataset by Olist (Kaggle) (Orders, customers, products, sellers, payments, and reviews)

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

All the 3 are famous Valid Datasets. All are to be considered in the Project.

Abstract

The rapid growth of e-commerce has generated large volumes of sales data, making accurate sales forecasting an essential task for businesses. Predicting future sales helps organizations optimize inventory management, reduce operational costs, improve customer satisfaction, and make informed business decisions. Traditional forecasting methods often struggle to capture complex patterns in sales data, whereas machine learning techniques can effectively analyze historical data and identify hidden trends. This project focuses on developing a machine learning-based sales forecasting system using historical e-commerce sales data. The project begins with data collection and preprocessing, including handling missing values, removing duplicate records, encoding categorical variables, and feature scaling. Exploratory Data Analysis (EDA) is then performed to understand customer purchasing behavior, seasonal trends, and the factors influencing sales. Several machine learning algorithms, including Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor, are implemented and compared to determine the most accurate forecasting model. The performance of these models is evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. Visualizations such as sales trend graphs, heatmaps, correlation matrices, and feature importance charts are used to present the analysis and

results. The expected outcome of this project is a reliable sales forecasting model that enables businesses to predict future demand accurately. The proposed system can support inventory planning, marketing strategies, pricing decisions, and overall business growth by providing data-driven insights for effective decision-making.

Data Analytics (EDA Flow)

Raw Dataset

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Dataset Overview

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Data Quality Assessment

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Descriptive Statistics

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Missing Values

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Duplicate Records

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Outlier Detection

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Univariate Analysis

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Bivariate Analysis

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Multivariate Analysis

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Correlation Analysis

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Seasonality & Trend Analysis

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Feature Engineering

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Feature Selection

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Machine Learning Models

Preprocessing

- Missing Value Handling
- Duplicate Removal
- Label Encoding / One-Hot Encoding
- Feature Scaling
- Date-Time Feature Extraction

- Train-Test Split

Machine Learning Models

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor
- LightGBM (Optional)

Visualization

- Sales Trend Graph
- Heatmap
- Correlation Matrix
- Monthly Sales Analysis
- Category-wise Sales
- Feature Importance
- Dashboard

Performance Evaluation

- MAE
- MSE
- RMSE
- R^2 Score

Recommended Research Pipeline

Benchmark Dataset

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Data Cleaning

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EDA

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Feature Engineering

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Feature Selection

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Train-Test Split

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Machine Learning Models

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Performance Evaluation

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Visualization Dashboard

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Comparative Analysis



Research Conclusions

Research Conclusion

This project predicts future e-commerce sales using historical sales and customer behavior data. It compares multiple machine learning models to identify the most accurate forecasting approach and helps improve inventory management, demand planning, pricing strategies, and business decision-making.

Batch details

Batch number:14

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